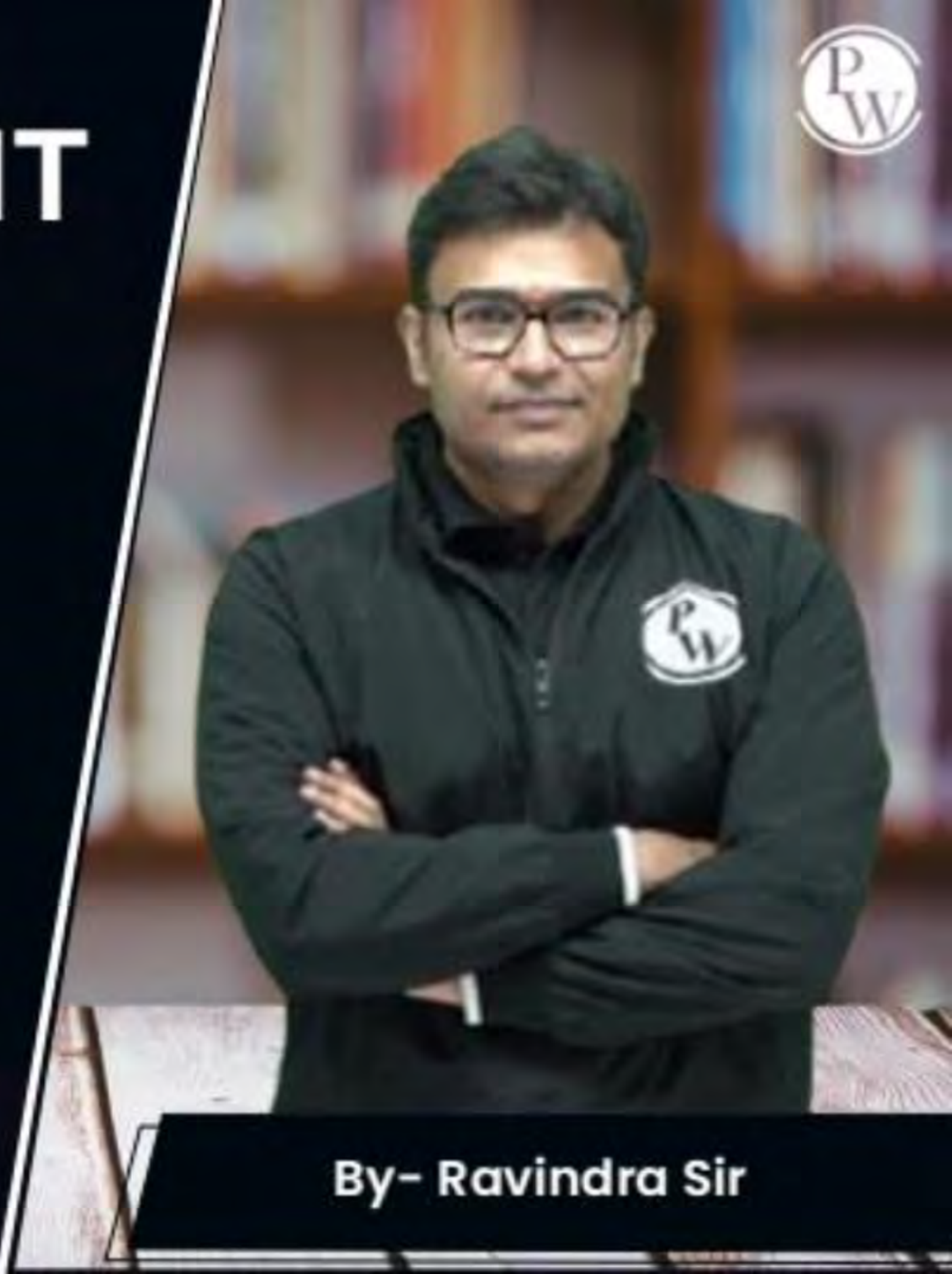
Computer Science & IT

COMPILER DESIGN
(CD)

Parsers

Lecture No. 6



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

First and follow

Topics to be Covered



- Topic
- Topic
- Topic

first and follow



Extra Ordinary Individuals: Stories to Ignite Student Motivation

Meghnad Saha (1893–1956) developed the Saha ionization equation, explaining how stellar atmospheres become ionized at high temperatures. Working at Calcutta University with limited resources, he applied thermodynamics and statistical mechanics to astrophysics, accurately predicting stellar spectra. Saha also founded institutions for nuclear and cosmic ray research in India and served as a member of the Planning Commission. His ability to bridge fundamental theory with observational astronomy laid groundwork for modern astrophysics and national scientific infrastructure.

Lesson: Applying basic principles creatively, even with scarce resources, can unlock new fields of study.



Extra Ordinary Individuals: Stories to Ignite Student Motivation

C. N. R. Rao (b. 1934) is one of the most cited chemists in the world. At the Indian Institute of Science, he pioneered research on transition metal oxides, unraveling properties of superconductors, ferroelectrics, and multiferroics. Rao founded the Jawaharlal Nehru Centre for Advanced Scientific Research and mentored countless researchers, fostering interdisciplinary collaboration. His lifelong dedication to materials chemistry and science policy has elevated India's research landscape and inspired generations of chemists to pursue bold, innovative investigations.

Lesson: Sustained passion for research and mentorship cultivates innovation and future scientific leaders.



Extra Ordinary Individuals: Stories to Ignite Student Motivation

Har Gobind Khorana (1922–2011) **decoded the genetic code and synthesized the first artificial gene. Born in rural Punjab, he earned a PhD at Cambridge and conducted pivotal experiments at MIT and the University of Wisconsin. Khorana's work demonstrated how nucleotide sequences direct protein synthesis, revolutionizing molecular biology. Awarded the Nobel Prize in 1968, he remained humble and dedicated to training young scientists, championing collaborative research across continents and laying the foundations for modern genetic engineering.**

Lesson: **Rigorous experimentation and global collaboration can transform our understanding of life's blueprint.**



Extra Ordinary Individuals: Stories to Ignite Student Motivation

Roddam Narasimha (b. 1933) was an eminent aerospace scientist who advanced India's space and defense programs. After completing his doctorate at Caltech, he returned to lead fluid dynamics and supersonic flight research at the National Aerospace Laboratories. Narasimha designed wind tunnels, developed turbulence models, and advised ISRO on launch vehicle aerodynamics. His strategic insights blended deep technical mastery with policy vision, propelling India toward independent spaceflight capabilities and inspiring a generation of engineers.

Lesson: Combining technical expertise with strategic vision accelerates national technological self-reliance.

	first	follow
$S \to ABCD$	$\{b, c\}$	$\{\$\}$
$A \to b \mid \epsilon$	$\{b, \epsilon\}$	$\{c\}$
$B \to c$	$\{c\}$	$\{d\}$
$C \to d$	$\{d\}$	$\{e\}$
$D \to e$	$\{e\}$	$\{\$\}$

$AB \subseteq D$

$ABCD$

$AB\underline{CD}$

$follow(A) = First(BCD)$

$S \to ABCD$

$follow(B) = First(CD)$

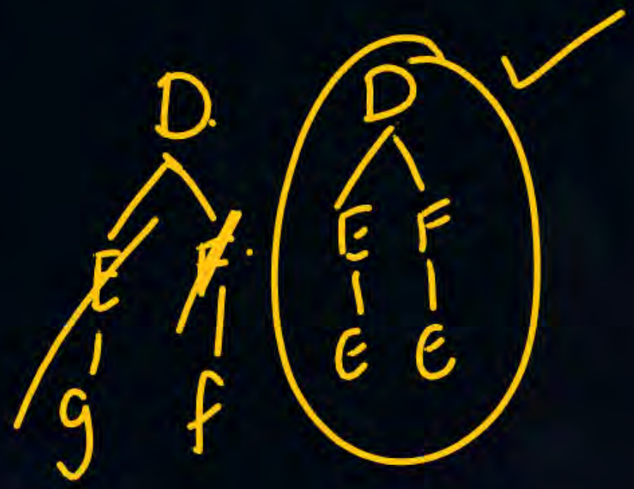
	First	follow
$S \to ABCDE$	$\{a, b, c\}$	$\{\$\}$
$A \to a/\epsilon$	$\{a, \epsilon\}$	$\{b, c\}$
$B \to b/\epsilon$	$\{b, \epsilon\}$	$\{c\}$
$C \to c$	$\{c\}$	$\{d, e, \$\}$
$D \to d/\epsilon$	$\{d, \epsilon\}$	$\{e, \$\}$
$E \to e/\epsilon$	$\{e, \epsilon\}$	$\{\$\}$

$S \to ABCDE$

$S \to ABCDE$

	First	follow
$S \to aBDh$	$\{a\}$	$\{\$\}$
$B \to cC$	$\{c\}$	$\{g, f, h\}$
$C \to bC / \epsilon$	$\{b, \epsilon\}$	$\{g, f, h\}$
$D \to EF$	$\{g, f, \epsilon\}$	$\{h\}$
$E \to g / \epsilon$	$\{g, \epsilon\}$	$\{f, h\}$
$F \to f / \epsilon$	$\{f, \epsilon\}$	$\{h\}$

$S\$$



Constructing of LL(1) parsing table:-

The rows in the table are given by non terminals and columns in the table are given by terminals

	a	b	c	$
S				
A				
B				

Algorithm:- For building LL(1) parsing table 'm'

1) For each production $A \to \alpha$ of the grammar, perform step 2 and 3
2) For each terminal a in First(α), add $A \to \alpha$ to m[S, a]
3) If 'ϵ' is in first(α) add $A \to \alpha$ to m[S, b] for each terminal 'b' in follow(A)

	First	follow
$E \to TE'$	{id, (}	{$,)}
$E' \to +TE' / \epsilon$	{+, ε}	{$,)}
$T \to FT'$	{id, (}	{+, $,)}
$T' \to *FT' / \epsilon$	{*, ε}	{+, $,)}
$F \to id / (E)$	{id, (}	{*, +, $,)}

$E \to TE'$

↓ Now

First (TE')

Fir (TE')

id (

$E' \to (+TE')$

a) Repeat b) NO

$E' \to \epsilon$

$T \to FT'$

First (FT')

$T' \to \epsilon$

	id	+	*	(	)	$
E	$E \to TE'$			$E \to TE'$		
E'		$E' \to +TE'$		$E' \to \epsilon$		$E' \to \epsilon$
T	$T \to FT'$			$T \to FT'$		
T'		$T' \to \epsilon$	$T' \to *FT'$		$T' \to \epsilon$	$T' \to \epsilon$
F	$F \to id$			$F \to (E)$		

$S \to (S) / \epsilon$

$S \to (S)$ ✓ ✓

$F((S)) = ($

$S \to \epsilon$

	first	follow
S	{(, ∈}	{), $}

LL(1)

	(	)	$
S	S→(S)	S→∈	S→∈

	first	follow
$E \to TE'$	{id, (}	{), $}
$E' \to +TE' / \epsilon$	{+, ϵ}	{), $}
$T \to FT'$	{id, (}	{+,), $}
$T' \to *FT' / \epsilon$	{*, ϵ}	{+,), $}
$F \to id / (E)$	{id, (}	{*, +,), $}

$E \to TE'$

First(TE')

id

(

	id	+	*	(	)	$
E	$E \to TE'$			$E \to TE'$		
E'						
T						
T'						
F						

	first	follow
$E \to TE'$	{id, (}	{), $}
$E' \to +TE' / \epsilon$	{+, ϵ}	{), $}
$T \to FT'$	{id, (}	{+,), $}
$T' \to *FT' / \epsilon$	{*, ϵ}	{+,), $}
$F \to id / (E)$	{id, (}	{*, +,), $}

$E' \to +TE'$

$+TE'$

$E' \to \epsilon$

	id	+	*	(	)	$
E	$E \to TE'$			$E \to TE'$		
E'		$E' \to +TE'$			$E' \to \epsilon$	$E' \to \epsilon$
T						
T'						
F						

	first	follow
$E \to TE'$	{id, (}	{), $}
$E' \to +TE' / \epsilon$	{+, ϵ}	{), $}
$T \to FT'$	{id, (}	{+,), $}
$T' \to *FT' / \epsilon$	{*, ϵ}	{+,), $}
$F \to id / (E)$	{id, (}	{*, +,), $}

$T \to (FT')$

First(FT')

id

(

	id	+	*	(	)	$
E	$E \to TE'$			$E \to TE'$		
E'		$E' \to +TE'$			$E' \to \epsilon$	$E' \to \epsilon$
T	$T \to FT'$			$T \to FT'$		
T'						
F						

	first	follow
$E \to TE'$	{id, (}	{), $}
$E' \to +TE' / \epsilon$	{+, ϵ}	{), $}
$T \to FT'$	{id, (}	{+,), $}
$T' \to *FT' / \epsilon$	{*, ϵ}	{+,), $}
$F \to id / (E)$	{id, (}	{*, +,), $}

$T' \to (*FT')$

$T' \to \epsilon$

	id	+	*	(	)	$
E	$E \to TE'$			$E \to TE'$		
E'		$E' \to +TE'$			$E' \to \epsilon$	$E' \to \epsilon$
T	$T \to FT'$			$T \to FT'$		
T'		$T \to \epsilon$	$T' \to *FT'$		$T \to \epsilon$	$T \to \epsilon$
F						

	first	follow
$E \to TE'$	{id, (}	{), $}
$E' \to +TE' / \epsilon$	{+, ϵ}	{), $}
$T \to FT'$	{id, (}	{+,), $}
$T' \to *FT' / \epsilon$	{*, ϵ}	{+,), $}
$F \to id / (E)$	{id, (}	{*, +,), $}

$T' \to (*FT')$

$T' \to \epsilon$

	id	+	*	(	)	$
E	$E \to TE'$			$E \to TE'$		
E'		$E' \to +TE'$			$E' \to \epsilon$	$E' \to \epsilon$
T	$T \to FT'$			$T \to FT'$		
T'		$T \to \epsilon$	$T' \to *FT'$		$T \to \epsilon$	$T \to \epsilon$
F						

	first	follow
$E \to TE'$	{id, (}	{), $}
$E' \to +TE' / \epsilon$	{+, ϵ}	{), $}
$T \to FT'$	{id, (}	{+,), $}
$T' \to *FT' / \epsilon$	{*, ϵ}	{+,), $}
$F \to id / (E)$	{id, (}	{*, +,), $}

$F \to id$

$F \to (E)$

	id	+	*	(	)	$
E	$E \to TE'$			$E \to TE'$		
E'		$E' \to +TE'$			$E' \to \epsilon$	$E' \to \epsilon$
T	$T \to FT'$			$T \to FT'$		
T'		$T' \to \epsilon$	$T' \to *FT'$		$T' \to \epsilon$	$T' \to \epsilon$
F	$F \to id$			$F \to (E)$		

$S \to AaAb / BbBa$

$A \to \epsilon$ ✓

$B \to \epsilon$ ✓

$S \to AaAb$

$S \to BbBa$

ϵ

$S \to (AaAb)$

	a	b ✓	$
S	$S \to AaAb$	$S \to BbBa$	
A	$A \to \epsilon$	$A \to \epsilon$	
B	$B \to \epsilon$	$B \to \epsilon$	

$S \to BbBa$

	fir	fol
$S \to AaAb$	a	$
$S \to BbBa$	b	$
$A \to \epsilon$	ϵ	a, b
$B \to \epsilon$	ϵ	a, b

THANK - YOU